Chapter 3

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Exercise 1.

Let $\phi : Y \to X$ be a nonconstant holomorphic map of Riemann surfaces, with X connected. Show that for all $x \in X$ we have

$$\sum_{y \in \phi^{-1}(x)} e_y = n,$$

where n is the cardinality of the fibers not containing branch points, and e_y is the ramification index at $y \in Y$.

Proof.
WLOG assume that Y is connected as well, as we can sum over each connected component.

If x does not lie below branch points of Y, then we are immediately done as $\sum_{y \in \phi^{-1}(x)} e_y = \sum_{[Y:X]} 1 = n$.

Now suppose that x has a branch point of Y. For each point y_i in the fiber of x, there exists a neighbourhood V_i and a complex chart $g_i : V_i \to \mathbb{C}$, $f_i : \phi(V_i) \to \mathbb{C}$ such that $(f_i \circ \phi \circ g_i^{-1})(z) = z^{e_y}$ when expressed in this chart. Hence, pick a x_0 that is in a sufficiently small (punctured) open neighbourhood of x such that all its fibers land in V_i. Then:

$$\begin{aligned} n = \#[\phi^{-1}(x_0)] &= \sum_i \#[\phi^{-1}(x_0) \cap V_i] \\ &= \sum_i \#[(f_i \circ \phi \circ g_i^{-1})(f_i(x_0))] \\ &= \sum_i \#\left[(z \mapsto z^{e_y})^{-1}(f_i(x_0))\right] \\ &= \sum_i e_y \\ &= \sum_{y \in \phi^{-1}(x)} e_y. \ \blacksquare \end{aligned}$$

Exercise 2.

Consider a nonconstant holomorphic map $\phi : P^1(\mathbb{C}) \to P^1(\mathbb{C})$.

(a) Show that there exists a unique rational function $f \in \mathbb{C}(t)$ such that $\phi = \phi_f$ as in the proof of Proposition 3.3.11.

Proof.
For existence, by Corollary 3.3.12. there exists an anti-isomorphism $\mathrm{Hom}(P^1(\mathbb{C}), P^1(\mathbb{C})) \cong \mathrm{Hom}(\mathcal{M}(P^1(C)), \mathcal{M}(P^1(C)))$ that sends $\phi \mapsto \phi^*$. Hence we are reduced to understanding $\mathbb{C}$-linear embeddings $\phi^* : \mathbb{C}(t) \hookrightarrow \mathbb{C}(t)$. But then this is entirely determined by the image of t; which is necessarily some rational function f. Hence $\phi(z) = \phi^*(z) = f(z)$.

For uniqueness, suppose that f, g are two such rational functions. Then $\frac{f}{g}$ is a rational function that is 1 everywhere except for the poles of f, g. By continuity $\frac{f}{g} = 1$ everywhere, so $f = g$. ■

(b) Relate the branch points of ϕ_f to zeros and poles of f, and determine the ramification indices.

Proof.

First suppose that $z_0 \in \mathbb{C} \setminus P(f)$. Pick a sufficiently small neighbourhood U of z_0 such that neither U, $V := f(U)$ contains infinity. We apply the appropriate linear transformation to U and V to center both at the origin:

$$\begin{tikzcd}
U \arrow[r, "f"] \arrow[d, "z - z_0"'] & V \arrow[d, "z - f(z_0)"] \\
U_0 \arrow[r, dashed] & V_0
\end{tikzcd}
\qquad
\begin{tikzcd}
z + z_0 \arrow[r, mapsto] & f(z+z_0) \arrow[d, mapsto] \\
z \arrow[u, mapsto] \arrow[r, dashed, mapsto] & f(z+z_0) - f(z_0)
\end{tikzcd}$$

then z_0 is a branch point if and only if the zero of $f(z + z_0) - f(z_0)$ at the origin is not simple; i.e. $\lim_{z \to 0} \frac{f(z+z_0)-f(z_0)}{z} = f'(z_0) = 0$. The ramification index then is exactly $1 + \mathrm{ord}_{z_0}(f')$.

Equally, when $z_0 \in P(f) \cup \{\infty\}$ we can apply the same logic but instead with the function $1/f(z)$, to get the following result: A point $p \in P^1(\mathbb{C})$ is a branch point if and only if it is a non simple zero of $f'(z)$ or $(1/f)'(z)$, with ramification index exactly equal to one plus its order as a pole or zero. ■

(c) Show that ϕ_f is a holomorphic automorphism of $P^1(\mathbb{C})$ if and only if $f = (at+b)(ct+d)^{-1}$ with some $a, b, c, d \in \mathbb{C}$ satisfying $ad - bc \neq 0$. [*Hint:* Observe that if ϕ_f is an automorphism, f can only have a single zero and pole, and these must be of order 1.]

Proof.
Let f be such an automorphism. Then it can necessarily only have a single zero and pole, which must be of order 1. Hence we can write $f(t) = \frac{at+b}{ct+d} = \frac{a}{c}\frac{t+b/a}{t+d/c}$. This function is nonconstant iff the pole and the zero do not cancel each other out; which is true iff $\frac{b}{a} \neq \frac{d}{c}$ iff $ad - bc \neq 0$.

Finally, we verify that such nonconstant f always define automorphisms of $P^1(\mathbb{C})$: For surjectivity, note that the image of f must be open by the Open Mapping Theorem. The image must also be compact, but as $P^1(\mathbb{C})$ is Hausdorff it must be closed. By the connectedness of $P^1(\mathbb{C})$ we are done. Injectivity follows from the fact that the (formal) inverse of f given by $f^{-1}(t) = \frac{dt-b}{-ct+a}$ can take on at most one value. ■

(d) Deduce that a holomorphic automorphism of $P^1(\mathbb{C})$ has one or two fixed points.

Proof.
By the previous result, let $f = \frac{at+b}{ct+d}$ be a holomorphic automorphism of $P^1(\mathbb{C})$. Then $f(t) = t$ reduces to a quadratic in t, such that f has one or two fixed points (depending on the root multiplicity of the resultant quadratic). ■

Exercise 3.

Let $Y \to X$ be a holomorphic map of compact Riemann surfaces, restricting to a cover $Y' \to X'$ outside the branch points.

(a) Show that the étale $\mathcal{M}(X)$-algebra $\mathcal{M}(Y)$ is isomorphic to a finite direct sum of copies of $\mathcal{M}(X)$ if and only if the cover $Y' \to X'$ is trivial.

Proof.

The given statement is not true if you do not assume X is connected: Take $X = P^1(\mathbb{C}) \sqcup P^1(\mathbb{C})$, and $Y = P^1(\mathbb{C}) \sqcup (P^1(\mathbb{C}) \sqcup P^1(\mathbb{C}))$. This is a trivial cover with $\mathcal{M}(X) \cong \mathbb{C}(t) \oplus \mathbb{C}(t)$ and $\mathcal{M}(Y) \cong \mathbb{C}(t) \oplus \mathbb{C}(t) \oplus \mathbb{C}(t)$. One has $\mathbb{C}$-dimension 2 and the other has 2, so $\mathcal{M}(Y)$ cannot be a free module over $\mathcal{M}(X)$.

Thus, for the rest of this question (including (b)) we will be assuming that X is connected.

As $\mathcal{M}(Y) \cong \prod \mathcal{M}(Y_i)$, WLOG assume that Y is connected such that we are reduced to the case of considering the extension $\mathcal{M}(Y) = \mathcal{M}(X)$. Note that by compactness there must be finitely many connected components. By the equivalence of categories in Theorem 3.3.7., the extension is trivial iff Y' is degree 1 cover of X', i.e. is X' itself. Hence $\mathcal{M}(Y)$ decomposes into a finite direct sum iff each connected component of $Y' \to X'$ is trivial, i.e. the cover is trivial. ■

We make the additional observation that the number of copies of $\mathcal{M}(X)$ is exactly equal to the degree of the covering. Hence the question statement can be strengthened to the following:

The étale $\mathcal{M}(X)$-algebra $\mathcal{M}(Y)$ is isomorphic to a direct sum of n copies of $\mathcal{M}(X)$ if and only if the cover $Y' \to X'$ is trivial of degree n.

This will be relevant in (b) below.

(b) Using Chapter 2, Exercise 2 give another proof of the fact that in the anti-equivalence of Theorem 3.3.7 Galois branched covers correspond to Galois extensions.

Proof.

We start with some finite extension $\mathcal{M}(Y)/\mathcal{M}(X)$. By the primitive element theorem, write $\mathcal{M}(Y) = \mathcal{M}(X)[\alpha]$ for some α. In the case where the extension is Galois, denote the Galois group by G.

First notice that this extension is Galois if and only if the minimal polynomial m_α splits completely in $\mathcal{M}(Y)$, as $\mathcal{M}(X)[\alpha] \cong \mathcal{M}(X)[t]/(m_\alpha(t))$ would be the splitting field of m_α.

Next we show that $\boldsymbol{\mathcal{M}(Y) \otimes_{\mathcal{M}(X)} \mathcal{M}(Y) \cong \bigoplus_{|G|} \mathcal{M}(Y)}$ **if and only if** $\boldsymbol{\mathcal{M}(Y)/\mathcal{M}(X)}$ **is Galois**: We have that $\mathcal{M}(Y) \otimes_{\mathcal{M}(X)} \mathcal{M}(Y) \cong \mathcal{M}(X)[t]/(m_\alpha) \otimes_{\mathcal{M}(X)} \mathcal{M}(Y) \cong \mathcal{M}(Y)[t]/(m_\alpha)$ as $\mathcal{M}(X)$-algebras. By the Chinese remainder theorem, $\mathcal{M}(Y)[t]/(m_\alpha)$ splits into a finite direct sum of quotients by the irreducible factors of m_α. Hence each component is isomorphic to $\mathcal{M}(X)$ if and only if the factor is linear; i.e. m_α splits completely. In particular, when this does happen, m_α must split into exactly $\deg m_\alpha = [\mathcal{M}(Y) : \mathcal{M}(X)] = |G|$ many factors.

Next we show that $\boldsymbol{\mathcal{M}(Y \times_X Y) = \mathcal{M}(Y) \otimes_{\mathcal{M}(X)} \mathcal{M}(Y)}$. By Theorem 3.3.7. we have that finite Galois branched covers are anti-equivalent to finite Galois extensions of $\mathcal{M}(X)$ via the $\mathcal{M}(-)$ functor. Notice that the fibered product over X is exactly the product in the slice category of spaces over X, and the tensor product is the coproduct in the category of rings/modules. But then $\mathcal{M}$ is a (contravariant) equivalence, so it commutes with the product to form the coproduct.

By the result of (a), $\boldsymbol{(Y \times_X Y)'}$ **is isomorphic to a degree** $\boldsymbol{|G|}$ **trivial cover if and only if** $\boldsymbol{\mathcal{M}(Y \times_X Y) \cong \bigoplus_{|G|} \mathcal{M}(Y)}$.

Next notice that $\boldsymbol{(Y \times_X Y)' = Y' \times_{X'} Y'}$, viewing both as subsets of $Y \times Y$. This step is straightforward, as the cover $(Y \times_X Y)' \to X'$ is obtained via removing all the points that shares fibers with a branch point.

By the result of Chapter 2 Exercise 2, $\boldsymbol{Y' \to X'}$ **is** $\boldsymbol{G}$**-Galois if and only if** $\boldsymbol{Y' \times_{X'} Y' \cong Y' \times G}$ **over** $\boldsymbol{Y'}$.

We have now proven every statement necessary for the question. To spell it out:

$$\begin{aligned}
\mathcal{M}(Y)/\mathcal{M}(X) \text{ Galois} &\iff \mathcal{M}(Y) \otimes_{\mathcal{M}(X)} \mathcal{M}(Y) \cong \textstyle\bigoplus_{|G|} \mathcal{M}(Y) \\
&\iff \mathcal{M}(Y \times_X Y) \cong \textstyle\bigoplus_{|G|} \mathcal{M}(Y) \\
&\iff (Y \times_X Y)' \text{ is isomorphic to a degree } |G| \text{ trivial cover} \\
&\iff Y' \times_{X'} Y' \cong Y' \times G \\
&\iff Y' \to X' \text{ is } G\text{-Galois. } \blacksquare
\end{aligned}$$

Exercise 4.

Let X be a compact connected Riemann surface, and $\phi : X \to P^1(\mathbb{C})$ a holomorphic map. Show that if ϕ is not an isomorphism, then its branch points cannot all lie above the same point of $P^1(\mathbb{C})$. [*Hint:* Use the Riemann–Hurwitz formula.]

Proof.
By the Riemann-Hurwitz formula,

$$2g_X - 2 = d(2g_Y - 2) + \sum_y (e_y - 1)$$

where the sum is taken over all the branch points of ϕ. Now suppose that $Y = P^1(\mathbb{C})$ and all the branch points lie above some common point x_0. Then

$$2g_X - 2 = -2d + \sum_{y \in \phi^{-1}(x_0)} e_y - \#\phi^{-1}(x_0)$$
$$2g_X - 2 = -2d + d - \#\phi^{-1}(x_0)$$

where $\sum_{y \in \phi^{-1}(x_0)} e_y = d$ by the result of Exercise 1. But then $2g_X - 2 = -d - \#\phi^{-1}(x_0) < -1 - 1 = -2$, such that $g_X < 0$. But then this is not possible. ■

Exercise 5.

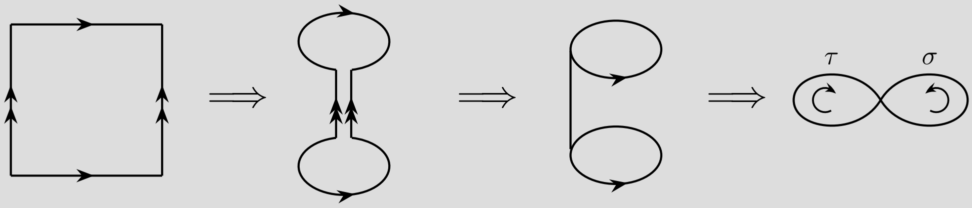
Let $n > 2$ be an even integer, and consider the dihedral group

$$D_n = \langle \sigma, \tau \mid \sigma^n = \tau^2 = 1, \sigma\tau = \tau\sigma^{n-1} \rangle$$

Show that every complex torus $X = \mathbb{C}/\Lambda$ has a Galois branched cover $Y \to X$ with group D_n with exactly n branch points, all lying above the same point of X.

Proof.
Fix some parallelogonal fundamental domain for the lattice Λ, and start by removing some point p in the interior of the parallelogram.

Then the punctured parallelogram is homotopic to the empty square, which we see to be homeomorphic to the figure eight as follows:



The fundamental group of the figure eight is $\langle \tau \rangle * \langle \sigma \rangle \cong \mathbb{Z} * \mathbb{Z}$. Hence we get a quotient $\pi_1(X, x) \to D_n$ (where $x \neq p$ is some arbitrary point) with $\tau \mapsto \tau, \sigma \mapsto \sigma$. By Theorem 2.3.7. this corresponds to a cover $Y' \to X \setminus \{p\}$, which is Galois with automorphism group D_n by Theorem 2.2.10.. By Proposition 3.2.6. this cover naturally extends to a Galois branched cover of compact Riemann surfaces $\phi : Y \to X$. It remains to determine the number of branch points.

By Proposition 3.2.10., all the points above p are branch points of the same ramification index e, and by the result of Exercise 1 we see that $\deg \phi = e \cdot \#\phi^{-1}(p)$ so it is sufficient to determine e.

For each branch point y lying above p, pick a local chart $(V, g) \xrightarrow{\phi} (U, f)$ such that ϕ is locally of the form $z \mapsto z^e$. By Example 2.4.12., we know that $\pi_1(U, p)$ is generated by a simple loop γ around p; and the monodromy action of γ is trivial if and only if it is a multiple of e. Hence e is exactly the smallest positive integer such that γ^e has the trivial action.

To that end, by following the homeomorphism depicted in the diagram, a counterclockwise loop around the square corresponds to the loop $\sigma\tau\sigma^{-1}\tau^{-1}$, which is equivalent to σ^{-2} under the quotient. Hence we are equipped to prove a stronger statement:

$$e = \operatorname{ord}(\sigma^{-2}) = \begin{cases} m & n = 2m \\ n & n = 2m+1 \end{cases}$$

and thus

$$\#\phi^{-1}(p) = \frac{2n}{e} = \begin{cases} 4 & n = 2m \\ 2 & n = 2m+1 \end{cases}. \qquad \blacksquare$$

Exercise 6.

Let $\phi : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ be a holomorphic map of complex tori.

(a) Show that ϕ has no branch points.

Proof.
Note that complex tori have genus equal to 1. Hence by the Riemann Hurwitz formula, $2-2 = d(2-2) + \sum_y (e_y - 1)$ and $\sum_y (e_y - 1) = 0$. This is only possible when there are no branch points. ■

(b) Denoting by $p : \mathbb{C} \to \mathbb{C}/\Lambda$, $p' : \mathbb{C} \to \mathbb{C}/\Lambda'$ the natural projections, show that there exists a holomorphic map $\tilde{\phi} : \mathbb{C} \to \mathbb{C}$ with $p' \circ \tilde{\phi} = \phi \circ p$.

Proof.
For visibility we denote $\tilde{\phi}$ instead by Φ.

Fix some $z_0 \in \mathbb{C}$. If such a Φ exists, we must necessarily have that $p'(\Phi(z_0))) = \phi(p(z_0))$, such that $\Phi(z_0)$ must be in the fiber of p' above $\phi(p(z_0))$. Fix some $w_0 \in p'^{-1}(\phi(p(z_0)))$.

$$\begin{tikzcd} \mathbb{C} \arrow[r, dashed, "\Phi"] \arrow[d, "p"'] & \mathbb{C} \arrow[d, "p'"] \\ \mathbb{C}/\Lambda \arrow[r, "\phi"'] & \mathbb{C}/\Lambda' \end{tikzcd}$$

As $\mathbb{C}$ is simply connected, it is the universal cover of $\mathbb{C}/\Lambda$ by Proposition 2.4.9.. Hence given w_0, by the lifting property of the universal cover there exists a unique lift Φ with $\Phi(z_0) = w_0$ that makes the above diagram commute.

To see that Φ is holomorphic, fix some $z_0 \in \mathbb{C}$. Then let U, V denote sufficiently small neighbourhoods of $p(z)$ and $\phi(p(z))$ such that both covers p, p' are trivialized. Let U_0, V_0 denote the unique sheet above U, V containing z_0 and $\Phi(z_0)$. Then this gives us an atlas on which Φ is locally homeomorphic to a holomorphic function. Thus Φ is itself necessarily holomorphic. ■

(c) Check that setting $\tilde{\phi}(\infty) = \infty$ defines an extension of $\tilde{\phi}$ to a holomorphic map $P^1(\mathbb{C}) \to P^1(\mathbb{C})$, and show that this map is in fact a holomorphic automorphism of $P^1(\mathbb{C})$. [*Hint:* Use Exercise 4.]

Proof.
To show that Φ extends holomorphically in the manner described, it suffices to show that there exists $k \geq 0$ such that $\lim_{z\to\infty} \frac{1}{z^k}\Phi(z) = 0$ exists; i.e. Φ is meromorphic at infinity. Take $k = 2$.

To that end, fix a basis $\{\omega_1, \omega_2\}$ and a fundamental domain D for Λ. Then every complex number z can be uniquely written of the form $z_0 + n\omega_1 + m\omega_2$ for $z_0 \in D$ and n, m integers. Now consider the function $\Phi(z) - \Phi(z + \omega_i)$ $(i = 1, 2)$. As $p' \circ \Phi = \phi \circ p$, by chasing the diagram in (b) we see that $p' \circ \Phi$ is a constant function. But then by the open mapping theorem and the discreteness of Λ' this is only possible when $\Phi(z) - \Phi(z + \omega_i) = c_i$ is constant. Putting everything together, we see that

$$\frac{1}{z^k}\Phi(z) = \frac{\Phi(z_0) + nc_1 + mc_2}{(z_0 + n\omega_1 + m\omega_2)^k}.$$

For z to tend to infinity, one of or both of $|n|, |m|$ must go to infinity as well. In the case where $|n| \to \infty$ only, we see that

$$\lim_{z\to\infty}\left|\frac{1}{z^2}\Phi(z)\right| = \lim_{|n|\to\infty} \frac{\left|\Phi(z_0)/n^2 + c_1/n + mc_2/n^2\right|}{\left|z_0/n + \omega_1 + m\omega_2/n\right|^2} \to 0\,.$$

The proof for the other two cases is analogous. So Φ indeed extends holomorphicially to $P^1(\mathbb{C})$.

By the result of (a), this extended function can only have branch points above infinity. Hence by the result of Exercise 4, ϕ is necessarily an isomorphism. ∎

(d) Establish a correspondence between holomorphic maps $\phi : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$ and pairs $(a, b) \in \mathbb{C}^2$ with $a \neq 0$ and $a\Lambda + b \subset \Lambda'$. [*Hint:* Use Exercise 2.]

Proof.
The statement of the question is incorrect. The correspondence should be between pairs $(a, b) \in \mathbb{C}^2$ with $a \neq 0$ and $a\Lambda \subseteq \Lambda'$. As we will see there is no condition on b.

By the result of Exercise 2(c), and part (c) above, isomorphisms $P^1(\mathbb{C}) \to P^1(\mathbb{C})$ are described exactly by (nondegenerate) Möbius transformations.

First, suppose we have some $\phi : \mathbb{C}/\Lambda \to \mathbb{C}/\Lambda'$. Write $\Phi(z) = \frac{az+b}{cz+d}$. As $\Phi(\infty) = \infty$ we have that $c = 0$, so Φ must be in fact an affine function of the form $az + b$ where $a \neq 0$. In particular, as $p' \circ \Phi = \phi \circ p$, we get that $p'(\Phi(z)) = p'(\Phi(z + \omega))$ for all $\omega \in \Lambda$; i.e. $\Phi(z) - \Phi(z + \omega) \in \Lambda'$. By substituting in $z = 0$ we get that $a\Lambda \subseteq \Lambda'$.

It remains to verify that such (a, b)-pairs always induce well-defined maps between the complex tori. To that end, it is sufficient to show that $\Phi(z) = az + b$ lifts through the quotients; i.e. for all $\omega \in \Lambda$, $\Phi(z) - \Phi(z + \omega) \in \Lambda'$. But then if $a\Lambda \subseteq \Lambda'$, we have that $\Phi(z) - \Phi(z + \omega) = -a\omega \in \Lambda'$. ∎